

An Ethics Model for Innovation: The PiE (Puzzle-solving in Ethics) Model

By Cansu Canca, Founder and Director, AI Ethics Lab

How should organizations put AI ethics and responsible AI into practice? Is the answer AI ethics principles and AI ethics boards or should everyone developing AI systems become experts in ethics? In this module, Canca presents the model developed and employed at AI Ethics Lab: The Puzzle-solving in Ethics (PiE) Model. The PiE Model is a comprehensive and structured practice framework for organizations to integrate ethics into their operations as they develop and deploy AI systems. The PiE Model aims to make ethics a robust and integral part of innovation and enhance innovation through ethical puzzle-solving.

Lecture Transcript

0:00 Hi, everyone. I'm Cansu Canca. I'm a philosopher and ethics consultant, and the Founder and Director of AI Ethics Lab. Today I will talk about how to integrate ethics into AI development and deployment. I will present to you my model, the PiE model, Puzzle-solving in Ethics, which is a comprehensive and structured way of putting ethics into action throughout AI lifecycle. So let me share my slides. And let's get started. Here, we go - the PiE model, Puzzle-solving in Ethics and AI innovation.

0:36 A little bit about myself, I'm a philosopher, I am trained in moral and political philosophy, and I specialize in Applied Ethics. I'm the Founder and Director of AI Ethics Lab, which is one of the first initiatives that focused exclusively on doing AI ethics research, with the goal of guiding AI practitioners in their ethical decision making as they are developing AI systems, as they are deploying AI systems. So from the very beginning, my goal with AI Ethics Lab was to put ethics into action. And in doing so, using all the tools, knowledge, existing research we have in Applied Ethics, applying them, using them in the context of AI, and for questions that we don't yet have the answers, doing AI ethics research, and adding that to this knowledge base. So that was our goal. And we've been doing this, I started the AI Ethics Lab 2017. So it's been going on, we've been working on this for over five years now. I also recently, last year, joined Northeastern University's newly established Institute for Experiential AI as their ethics lead and research associate professor. Here, the work at Northeastern University is connecting directly with the work that I've been doing at AI Ethics Lab. The Institute's goal is to utilize the academic resources, the knowledge, the methodologies, and bringing that to the, to solve the problems that industry is facing. So my work is related to responsible AI and working with the companies, institutions, companies, again practitioners, to solve their responsible AI related questions, issues, putting in place systems, frameworks, so really completely connecting with my work that I have been in AI Ethics Lab.

2:44 And finally, I am also serving as AI ethics and governance expert consultant to the UN, specifically for a project, a joint project between Interpol and United Nations Interregional Crime and Justice Research Institute to

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build a toolkit for Responsible AI innovation for law enforcement. So there, again, very connected to my other positions. What we are doing is using the AI ethics, responsible AI tools and knowledge that we have, and applying that this time for the purposes of law enforcement in their development of AI systems and in their deployment of AI systems, to guide them to use AI responsibly. Before I started working on AI, I worked for a very long time in ethics and health, over a decade, specifically on population level bioethics. So I held several different positions. Some of them I listed here. So I was at the Faculty of University of Hong Kong Medical School teaching medical ethics. I did research on Bioethics and Health Policy at Harvard Law School after my PhD. And during my PhD, I did research on Bioethics at World Health Organization. So the work that I do, that I did, in the context of ethics and health in bioethics was related to health policy, resource allocation, research ethics, so all of them about the ethical aspects of, within healthcare, all the questions that we face that are ethically relevant. And that connects actually very well with the work I do right now. Because bioethics, especially population level bioethics, is where we have the most well established literature and work in Applied Ethics, which is extremely relevant as we are working with AI ethics as well.

4:57 I am also told to say a little bit about my geographic background. So I'm Turkish, I am shooting this video in a little town called Datça facing the Mediterranean Sea. I am born and raised in Turkey, I did my first degrees here. I did my PhD in Singapore, National University of Singapore, and my postdoctoral research at Harvard Law School. I studied and practiced ethics in many countries. So Singapore, Hong Kong, Japan, Turkey, of course, Switzerland, and the United States, which gives me pretty good practical experience in different cultures in applying ethics, in practicing ethics in different cultures, as well as the theoretical background, because I did study different types outside of Western philosophy, also Asian philosophy, and Islamic philosophy as well. So that is all about myself.

6:00 So now let's take a step back and look at the AI ethics background, history. So I like showing this graph, because I think it illustrates really well how AI ethics, that conversation has been trending upwards in the last five years. So this is the 10 year graph of Google Trends. There are, there are different ways that you can search for the similar idea. So you can search 'ethics in AI', 'ethics of AI', I used the word 'AI ethics', but you're gonna get a very similar graph in all cases. And as you can see, until 2017, mid-2017, there is not much of, the AI ethics is not getting a lot of interest. What changes in 2017 is Facebook and Cambridge Analytica scandals have happened. And this is very typical for ethics because, unfortunately, ethics gets attention when something goes really wrong. Same thing happened in medical ethics. Same thing happened with research ethics, you know, research ethics really came about after the medical experimentation on humans in World War Two, and after World War Two. And only then research ethics became a thing. Medical Ethics in a similar way. And environmental ethics, same same mentality. As things go wrong, when we see something, really, something really tells us that people have been neglecting the ethical aspect of their work, then we turn to ethics and ask, well, how can we put ethics in practice within that domain?

7:50 So that's what happened in AI ethics as well. I started AI Ethics Lab right before the trend started peaking. So this was in 2017, and at that point, it was right before Facebook and Cambridge Analytica scandal became got the attention that it eventually got. So it was right before that. And the reason why I started AI Ethics Lab was actually very practical, because at that point, nobody really was talking about AI ethics. I mean, of course, there are always people who are working on niche topics, but industry was not interested. Academia was not interested in really looking at this, the ethical questions, looking at it from the Applied Ethics perspective. How we can

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develop tools, how we can put the ethical, how we can put ethics into AI development, AI lifecycle. This was not a question of interest. So as a result, in the absence of any, any, any real opportunities to do this, I started AI Ethics Lab, which was very timely, which turned out to be very timely, because then the scandals happened, and then people realized more and more things were going wrong. And now we are really paying attention.

9:16 It's more, it's even more recent in responsible AI, because responsible AI really was not a thing. So AI ethics, you can say well, we always had machine ethics, we had information ethics, engineering ethics, so we were maybe not using the AI ethics but we were going around the topic until we hit AI ethics. Responsible AI is really a phrase that we started using once AI ethics became interesting within the industry. I suppose the reason is because ethics sounds vague and elusive, and the industry, the practitioners, preferred something a little bit more clearer, a little bit more tangible, I suppose that's the reason why we are using the responsible AI terminology. I'm gonna get to that, but I think there is a good distinction and good way of distinguishing AI ethics and responsible AI in a useful and meaningful manner as well.

10:18 So now, looking at this trend, I don't, it should not come as a surprise that we are talking about AI ethics and responsible AI, but there is no clear path in practice. So, there is no established, standard way of doing AI ethics. As you are developing AI systems or as you decide to use an AI system in an organization, there is no clear practice to say okay, here is how we add ethics, how we make sure that the ethical aspects are checked, controlled, sorted. We are in a time where every organization is in a way experimenting on what is the best way to go about it. And within those experiments, I think two approaches are emerging, have emerged. One is the Research Ethics Model. And the other one is the 'Brute Force' Approach that I call. And I am going to explain what they are, as well as why. And I'm going to argue why they are inadequate, and ill fit to answer the issues that we face in AI ethics. And I will argue that we need a different model, we need a model that allows ethics to be a part of the innovation cycle without losing its space to do proper, robust ethical work. So we don't want to, in a way dumb down ethics to insert it as a checklist. But we want to make sure that it is integrated, and it has opportunities to look in-depth into ethical issues that matter. So this is the summary slide.

12:14 Now, we are going to start by taking one more step back. So I want to get us on the same page on what do we talk about AI ethics and responsible AI when we talk about it. So what is it really that we are discussing here? Ethics is a concept that we are all very familiar with, which comes with its blessings and its curses. So I think the blessing is that we all sort of have a, it's ethical, the question of ethics is not new to any of us. We know why it matters. But the downside of being so familiar with it is that we tend to talk past each other often because we might have different ideas of when we say ethics, different things can come to our minds. So let's avoid that problem and clarify things before we go forward.

13:10 So, in AI ethics, we are, we can divide the scope of AI ethics in several ways, this is how I do it. So I would say that there are three main components of AI ethics and responsible AI. One is the product itself, the content, the second one is the process. And the last one is the implementation. In content, what we want to do is we want to make sure that the technologies that we are developing and designing are ethical. So I think by now, most of us are aware that technologies are not value neutral, there is no such thing as a perfectly objective, and free from all value judgment technology. Most of the time, we have, we embed value judgments into the development process. It can start with the purpose of the technology that we are developing. It might be, there will be many, many decisions about which data to use, what proxy to use in the model, what model to use, and how to,

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for whom to create the tool, and on whom to use the tool. All of these involve value judgments. So we want to make sure that any value judgment, any ethical decision that goes into the product is a justified decision. The second part is about the process. Developing AI systems, developing a new technology means that there's a research and development phase. And many times, oftentimes, this research and development phase involves engagement, interaction with humans, or using individual, using human information. So, as we are engaging with this research and development phases that can result in harm on people, on societies, on environment, we want to make sure that the process, the R&D process, runs ethically. And finally, even if you have been, even if the process was great, and the product has all the ethical judgments evaluated before they are embedded into the product, we still have the implementation part, because we want to make sure that the technology that we created, that we developed, is used, deployed ethically.

15:43 So here, we have to understand that a lot of the technologies and especially when it comes to AI systems, a lot of the technologies, all are multipurpose technologies. We might be developing it for one thing, and it is useful for things that we have sometimes not even thought about initially, which is not an excuse, because we have the as people who are engaged with the development of a technology, we have an obligation to try to foresee potential uses. So an example that comes to my mind right now is how the voice assistants like Siri or Google Assistant, they were really built to have our engagement with the operating system in a more smooth and seamless manner. So instead of typing, we are just gonna ask Siri, can you show me my latest emails or things like that, you know, like simple engagements, tell me the weather, tell me if I have an appointment coming up. But what happened is people started telling their secrets to these voice assistants, they started sharing, feeling lonely, they started telling their vulnerabilities and in their vulnerable times, how these systems react really matters. So for example, if a teenager is talking about suicide, we don't want the voice assistants to encourage them, we want them to, we want them, even though this does not this was not the original purpose, we want them to be able to guide them to help. And that is exactly what happened in the months, we realized that the voice assistants can be encouraging almost almost encouraging the bad thoughts as the person is sharing. The model was changed and a helpline was introduced. So now if you start talking about triggering thoughts, once the system actually tries to provide help. So a harm that we have not foreseen before. Now, we are dealing with it in a way in from the design perspective.

18:12 So this is just one implementation. Of course, there are many ways and when I say implementation, please think about like, think about individual use, like a person using the tool, but also think about organizational use, a police force using an AI system, a hospital system, healthcare system using an AI system. So there are also these organizational criminal justice system using AI tools. There are the organizational ways, aspects of deployment which matter massively. And we have to foresee them and we have to deal with them within the AI ethics and responsible AI as well. So we can say that the scope of AI ethics and responsible AI goes from really the conception of the research project, all the way through research, development, design, deployment, commercialization and even updating because a lot of these systems are meant to gather data, even after they are released and learn from that data, which means that there are many things that can go wrong after they are released, after the product is finished and let's say shipped. And we focus on every aspect of AI systems. Ethical questions arise related to data related to algorithms, the tools, the tool itself, the user interface, and how we engage, how humans engage with it. So the practice, human computer interaction.

19:41 Here are some examples. There are many examples that we can give at this point, but just to give you the sense of the scope as with some examples, so we start, I put just four of them for example. So one of them, the

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first one on the far left, is the 'gaydar study' from Stanford, which meant to determine who is gay by looking through facial recognition, looking at the faces. The problem with this research was obviously it started from the idea of the research and the risks that it contained versus the potential benefits. So here the risks are massive, and the potential benefit is relatively very small. In addition to that, of course, there is a scientific question. The scientific validity, looking at individual faces, assuming facial features show sexual orientation is, of course, scientifically, a hugely problematic starting point. The second example that I have here is about facial recognition tools. So this was a study that came out from MIT, and that showed that the commercially available in use facial recognition technologies had much higher error rate for dark, dark skinned female faces in comparison to light skinned male faces. So the difference was that for light skinned males, man, the error rate was 0.8%. In comparison, for dark skinned women, the error rate was 34.7%. The reason it was the data, the fact that these facial recognition tools learned from the data sets that were, that contained many light skinned male faces, and not as many dark skinned faces or female faces.

21:55 The third example that I have here is from healthcare, healthcare and the risk assessment tools used in healthcare system. So the one in question here is a risk assessment tool that determines who needs extra care. In order to do that, the algorithm uses healthcare spendings as a proxy for health needs. The problem here was that for black communities, there is less money spent on black communities, which makes the algorithm assume that black patients are healthier than white patients, even though they are equally sick. So the problem here really came from not realizing that the spending is already, there is already a discrimination, a disadvantage, for black patients. And if you use that as a proxy, you are basically taking that existing unfairness and implementing it into an AI system to perpetrate it. And finally, the last example I have is about Facebook and YouTube platforms, and their search and search function. So here, it was realized that when you want to learn about vaccination from Facebook, search, vaccination related information on Facebook or YouTube, so you start typing. These platforms suggest immediately anti-vaxxer, anti vaccination pages content to you. And it's knowing that in many parts of the world, Facebook is the source of information for people. So instead of using normal search engines, many people, many parts in the world, people use Facebook to learn things. This creates a distortion in information and a distortion on how people access knowledge. And as a result, what kind of decisions they make and on what they base their decisions. So this was one can say that here we have an issue related to how the user interface, how the system is structured.

24:30 Just to list a few of the common AI ethics issues. Again, this is not an exhaustive list. These are just the things that we face we face most often: unjustified bias, violation of privacy, lack of transparency, manipulation, lack of accountability and problem of accuracy are issues that we see very often in AI systems. And as I said, taking steps still back, we are still going backwards to get the big picture. So, the core question of ethics, of AI ethics, but of ethics is what is the right, what is the good thing to do? So when we look at these these cases, what we are asking really is, well, we understand what is wrong, but what is the right thing to do? How do we fix these problems? And what is the right and fair system to implement?

25:27 So, what we are asking is, what should one do? That one can be the developer, can be the deployer, can be the policymaker, but we are really asking: What should one do as one is, as we are developing AI systems, using AI systems and even regulating AI systems. And in ethics, we are going about answering this question in a systematic and analytic way. And what we want to do is to determine the right action or a set of justified actions. So, you will hear many times that in ethics, there is no right answer. Well, that's wrong. In many cases, there are,

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there is a right answer in ethics. And in many cases, we might be able to, we not be might not be able to say here is the answer, but we will still be able to say here is a set of justifiable answers and the rest is just wrong. If you do anything else, you're doing something wrong. So ethics is action guiding, and it is logical, analytic, consistent. It is not just tastes or opinion or feelings or beliefs or the legal structure or the societal norms. We want to, I want to make sure that when we talk about ethics, you don't assume that we are just doing reflection for the sake of reflection. But we are really going about trying to find the right answer. And we use theories and reasoning tools that are provided from philosophy, from the discipline, to get to that answer.

27:14 So now that I hope we are all on the same page about what is ethics, what this AI ethics, what is the scope of AI ethics, let's get back to those two models that I mentioned at the very beginning. So, as we are trying to figure out how to put AI ethics into practice, as I mentioned, two approaches have emerged.

27:35 The first one is the one that is most obvious, the research ethics model. Why? Because research ethics is what we are most familiar with in practice. Research Ethics is well established, it has its practical structure. And in this practical structure, what we have is a research ethics board. It can be called institutional review boards in the US or research ethics committees in Europe. So we basically have this board that reviews research before it starts, and determines whether or not it is ethically permissible, and whether or not it should be allowed to proceed. And in doing so, this board uses principles, ethics principles. It really relies on ethics principles, so the principles are guiding the board's decision. And why do we think that this model was adopted by the AI ethics, by the AI community? Well, starting from 2015, and starting from 2015, there has been a surge of AI ethics principles published around the world. Private companies, governments, intergovernmental agencies, research organizations, universities, many organizations publish their AI ethics principles. So from 4 documents we have here on 2015. By the time we got to 2021, we have over 100 documents, suggesting AI ethics principles. And this is a toolkit that we created at AI Ethics Lab to understand what's going on, because this was a very interesting phenomenon.

29:27 So, in research ethics, we have principles, those principles have been established in the 1970s. There are, they are extracted from ethical theories namely, they are: respect for autonomy, respect for individual autonomy, prevention of harm and justice. So, these are the three main principles. And these principles are used and different instruments are utilized, are put forth such as informed consent, or risk assessment tools, in order to meet those principles. But everyone publishing their own principles is a little bit different, a bit of a different phenomenon. So we created this toolkit to understand the trends. So you can see the AI ethics principles here by the, on the map as a heatmap. So the darker countries are the ones that have more publications coming out. The pins indicate different types of organizations, you can sort it chronologically. We also have other functions in the toolkit that you don't see here, you can see you can have the summaries of the principles, you can compare them, there's some analysis provided, and so on so forth, because we really thought that we need to understand what's going on with AI ethics principles. And as organizations publish their AI ethics principles, most of them also put together an AI ethics board. Well, that is very similar to research ethics model.

31:07 What is wrong with research ethics model? Well, the first thing is that, remember that lifecycle that I showed, the scope that I showed about AI ethics? Well, research ethics model looks at research before research starts. And that's really it, there is no other, the board is not involved after the research starts. If there are going to be changes in the research, then you have to make those changes and send for approval to the board before you

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implement those changes as well. Clearly, one obvious problem is that well, this is just not sufficient. Like when are we going to deal with the questions that arise in the development? How are you going to deal with the questions that arise in design or deployment? There is no way that a research ethics board can do this. We can also imagine, by the way, just to put things into perspective, for example, Google search engine in just 2020 had over 600,000 changes implemented. So there were so many, over 600,000 experiments, in just one year, and they resulted in 4000, more than 4500 changes in the search function. Can you imagine trying to get pre approval for all of these changes?

32:34 So, if you imagine extending this expanding the scope of board, board approval, throughout this the life cycle, we are talking about an amazing bottleneck and a very unhappy board, I suppose, because they have to look at new changes all the time. So the model is just not fitting the way that AI system, AI development works. And also, there are other problems with this board structure. So the board relies on principles as I said, but anyone who works on principles knows that principles are guiding to a certain degree only. When you have complex questions, principles serve as a reminder of what to take into account but they don't, they are not a complete decision making tool. Precisely because they are driven from different theories and those theories conflict with each other. So it is, it is logically impossible for the principles to be always in alignment, which means that prints a board that relies on principles can only deal with questions that are not complex. And typically, the research ethics boards do not have to have ethics experts and do not have ethics experts on board. Which is a huge problem, obviously, because that means that once we are at the limit of principles, there is no more there is no more expertise in the board to make decision to help with decision making or analysis. We can say that, okay, we'll put the experts into these boards. Well, we are still having a problem because the way that the board operates is as I said, it's on pre approval, it's authoritative and it's distant. So, it is not a collaborative function. And understanding how AI systems develop and all the different questions that arise on in quite high frequency. A board cannot look at all of these questions and should not look at all of these questions in this distant manner. It is inefficient and it is very ill fated.

35:03 If that's not it, then what is the other option? So, what? The second approach that emerged, I think somewhat organically, as we were so rightly, so concerned about the ethical issues, but didn't know how to solve them, who should solve them and how should we have this expertise integrated into the development cycle? What happened is we got the industry, the practitioners adopted this 'brute force' approach. So basically, everybody's asking all of these questions right, like, what is the impact of AI systems on societies? What is the right thing to do, the fair thing to do in deploying an AI system? How do we, what are the technical aspects? How can we develop AI systems that are fair or that are empowering? And what is, what are the best policies and regulations for the ethical development or deployment of AI systems? So we are all asking these questions at the same time, as almost as like a brute force approach. So like with it is as if we asked them all at the same time with sufficient rigor, we will get the answers. Well, the problem of this is that each one of these questions require different expertise, different methodology in answering them, we need to utilize different methodologies, different background knowledge, different tools. A social scientists' tools differ greatly from the philosopher's tools, which differs greatly from applied engineers' tools, or a policy makers' tools. And the knowledge set of knowledge, that is not that is pre AI, you know, not even about AI but how do let's say we are talking about social networks? Well, how does it impact? How can I make sure that a social network is good for society has so many moving parts answering this question has so many moving parts?

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37:14 In order to understand this, we have to understand how do societies interact. So that comes from pre AI, and comes from social sciences. And then we have the technical aspects, you know, what can we do? What are the types of changes that we can implement on a social network, on a platform and to tie them all together? What are the value judgments we want to make? And if we are asking all of these questions at the same time, without, with disregarding our own expertise and our own knowledge base, we are really obscuring the value judgments, we are the result is usually, we are either misinformed because we did not listen to the lived experiences and the data that comes from social sciences, or we are very rigid, because we don't actually understand the concepts that are provided from ethics, applied ethics. And we may not even understand if we don't listen to the you know, applied aspect, applied engineering aspect or design aspect, we don't even know how can we create solutions? And not just look at is it ethical or is it not ethical? But rather how can we make this ethically better, ethically permissible or even ethically good?

38:38 So this is inefficient, this is wasting expert. This is wasting expert expert's time and losing the knowledge bases that we already have in many disciplines. And instead, I think here, we can talk about how responsible AI is a little different from AI ethics. You know, I mentioned at the beginning. I think, so, AI ethics is a part of philosophy. So AI ethics is really asking this question, among other questions related to AI systems. It's asking this question about the value question. So what is the right thing to do? What is the good thing to do? What is the fair, just thing to do in developing an AI system or deploying an AI system? So that is where AI ethics fits. It fits within philosophy. But responsible AI, I would argue is a broader way of thinking about how we can implement these value judgments. And to do this, we have to have different disciplines really collaborating with each other in a structured manner. So we don't want to lose any of these expertise. We want to make sure that there is space and opportunity to answer these questions in a robust manner, and to put them together for application, for implementation, and that is the implementation model that we need to create.

40:15 So now I am going to show you the puzzle-solving in ethics model, which focuses mostly on the ethics role, but allows, hopefully, as I will show, as you will see, it allows for these collaborations among different disciplines. It actually insists on those collaborations among different disciplines. Now, here's the basic structure of the PiE model, the puzzle-solving in ethics model, so, I developed this model in, the first iteration was in 2018, while when the discussion was really stuck on research ethics, and the brute force approach was not even yet there. And we have been since not just developing this model further, but what we have been doing is to add more components. So I will tell you, it will become clearer as I go through the model. And here's how we are going to go through the model, we are going to start from analysis down in the right hand side, and we are going to go over to the Strategy from process, go to playbook, go to people and come down to training. And finally, we are going to look at the roadmap that is underneath it all. So let's start with the analysis. First of all, why do we start with the analysis? Well, because everything we do is really to make sure that the analysis section is done properly and in a way that is integrated seamlessly and efficiently integrated into the innovation. So our whole purpose in using all of these other components is to keep the analysis robust and efficient.

42:15 So what is AI ethics analysis? So in AI ethics analysis, there are three main things that we want to do. The first one is we want to gather the information, we want to understand the domain and we want to understand the technology. So think about, for example, a risk assessment tool for healthcare, which was one of our examples earlier. So to understand the risk ethical landscape of this project. So let's say our project is to develop, is developing risk assessment tools for healthcare services. So to understand the ethical landscape that this project is

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now landing on, we want to understand the ethical questions within healthcare, existing ethical questions within healthcare. But we also want to understand existing ethical questions about risk assessment tools. So this information gathering part is really where we are learning about from other disciplines as ethicists, where we are learning about from other disciplines. We want to learn from, from the societal aspects, and we want to learn the technical aspects, and we want to understand the ethics, risks, and opportunities. And when you talk about ethics risks, I think everyone understands it, you know, what is the ethical risk that this project can introduce? But there is also the opportunity, let's not forget about that one, which is when we know what existing ethical issues are out there, we can also see if this project can solve some of those ethical issues, because, you know, our goal in ethics is not just 'oh, like, let's make sure that we can make these these tools as harmless as possible'. No, we want to create a good world. That's our basic goal. We have a very ambitious goal.

44:05 And for that goal, we want to make sure that the ethical risks in the world can be, we can find the ways we don't miss the opportunities to solve the existing ethical problems in the world. And we created a tool called The Box and we use that to do the ethics risks and opportunity assessments. Remember, I mentioned that we developed the PiE model, I developed the PiE model in 2018 but we've been adding components. So this is one of the components that we created, which has been, which we are also developing as we do more research and it has proved to be very useful. So we use the principles, those famous principles, but we put them in a structure where we have, we can see the. We put them in a structure where the tool helps us visually see what are the ethical weaknesses and the strengths of the project. And now that once we do that, we have an understanding of what to look out for what we need to what are the ethical risks that we need to eliminate or if not, mitigate? What are the safeguards maybe we need to put in place. But, you know, that's not like, we don't just like, ship off the evaluation. And that's it.

45:34 But what we want to do is to now come together with the developers and designers, or if this is a deployment case, with the deployers, and make sure that we solve these problems, we actually help them solve this. So we work together with the practitioners to put in place changes, modifications, revisions, through troubleshooting sessions, through ethics by design sessions, to implement these changes, to help them implement these changes so that the analysis is not just a document, but it actually is now integrated into the product. And this runs out throughout the whole project phase. So again, then as new questions arise, we update our risk analysis. And we continue with the troubleshooting and ethics by design sessions to make sure that these ethical issues are resolved or at least mitigated or at least noted. Even that is important, in the cases where there is an ethical weakness that we cannot mitigate, at least it's important to make sure that it is noted.

46:57 So analysis is not, should not happen. from scratch. Like, we should not start the analysis process from the scratch every single time there's a there's a new project. We don't want to rediscover how to do ethics analysis and how to integrate ethics analysis, what we want to have is an AI ethics strategy that includes a process of how to integrate AI ethics analysis into projects. So here, what we want to have in the process section is a workflow for every relevant department. Obviously, for innovation department, for example, but even for departments such as procurement department that buys AI systems, what are the ethical checkpoints? When should they ask which questions? And what kind of answers should they look for? So we want to have a process a set process that makes sure that ethics is a longitudinal, integrated part of the operation. And we also want to make sure that the decisions that are made, the judgments that are made during the analysis phase is not forgotten. Because they are, they serve as precedents for future decisions. Basically, understanding our reasoning mechanisms, understanding our value

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judgments, helps us deal with new cases. But it also helps an organization to keep a coherent value judgment. So we don't want an organization that is almost like that almost has like a split personality. We don't want an organization sometimes prioritizing user privacy, but other times, other times in various in almost exact same cases, just disregarding user privacy. We want to make sure that: one we are doing as we are making the best decisions from the ethics perspectives; and two whatever values that we are holding, not higher, we keep it consistent.

49:23 So recording the decisions. Keeping track of what's happening in analysis allows us to create this coherent system, which will guide the practitioners as well as the ethicists in the organization. And where did those records go? Well, they go to the playbook. A playbook consists of everything, every guiding material that an organization has for ethical decision making. And that goes for the decisions that the developers or designers are making, or the decisions that deployers are making, or the decisions that management and leadership is making. So we want to make sure that the playbook is as comprehensive and as guiding as possible. And we want to make sure that using the playbook is efficient. I keep emphasizing efficiency because well, because first of all, efficiency is also a very important, ethically very important aspect. You know, we don't want to lose resources that could be better utilized. So we are from an ethics perspective as well, we want efficiency. And I know also that practitioners are concerned about efficiency. So that's, that's one of the other, that's the other reason why I emphasize that I suppose.

50:44 So in the playbook, we have the principles, because principles turn out to be quite useful in helping practitioners make decisions on non-complex ethical cases, but also help them recognize complex ethical cases. So principles can be put in a format, which we do using The Box tool that I mentioned in the analysis. Principles can be put in a format that are operational, and that are a useful too. And we want to create tools, because we don't want, as I mentioned earlier, the goal here is not to turn every individual into an ethics expert reading tons of pages, but we want to make sure that every individual has the tools they need to do the ethical checks that they are tasked to do. And things are never clear until you see them in action. So use cases are extremely important to see how principles work, how tools really can be used, and how things come into play in a use case. And these use cases should also contain the decisions that are the actual decisions that are made in the analysis. Remember this feedback loop, where we have the analysis decisions recorded, they should feed back to the, to the playbook as use cases. And here, it's also important to keep in mind that we are really talking about a system where there is always a learning feedback loop going on. It could be the case that we, no matter how well we thought about the these these guiding principles or use cases and that the details of the playbook, it might turn out turn out that we didn't consider an outlier case, or a case that really showed us how our values might change given, in certain situations. So the playbook is you should consider the playbook as a living document that learns from your analysis. And you should consider the analysis guided by principles and the playbook, but also has the liberty to make a case if something seems like it is not properly covered in the playbook.

53:05 And finally, who's gonna do all of this work? Well, the ethics people, as I call them, is hope. So I think we all would like, as we are going through this, we're all imagining there's an ethics team who has ethics experts, yes, but that is exactly what we need. Because as I mentioned, just like you cannot do, you cannot create AI systems without people who are experts in building AI systems, or you cannot design great products without designers, or you cannot make do without a legal team, you need an ethics team who have experts in ethics. But as I mentioned, in the research ethics model, we also don't want to create a structure where the one board is creating a bottleneck, and also driving themselves crazy trying to make all the decisions in a gigantic organization, in some cases. What

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we instead want: an ethics network that is spread out throughout the whole organization. So here, what we mean is that in different layers of the organization, we will have different individuals tasked with, designated with different levels of ethics decision making. So in some cases, it can be an individual in an innovation team who just knows the playbook really well, who knows how to use the tools. So in when a question arises, that person can consult the playbook, use the tools and make most of the ethics decisions, because most ethics decisions are honestly the ones that are a lot of the ethical errors come from not realizing that there's an ethical problem, not being fully aware of the ethics risks, and not realizing the, not asking the question about what is the value added by this product and what is the risk created by this product.

55:05 So, once we have individuals who can sort of do these ethics checkups, we cover a lot of the bases, but those individuals should be skilled in making those checks. And they should also be skilled in recognizing when the question goes beyond the guiding tools. And when it should be escalated to the experts at the, in the ethics team. That will, that will allow the ethics team to remain as a useful resource for the organization without being overwhelmed with every little question. And some questions are about the organization's character, purpose, goals. And here, we want to have more than the ethics expertise, what we want to have is sort of like an Ethics Strategy Board that aligns the ethical policy decisions, if you will, with the, you know, the company's overall goals, overall, overall path. So we need all of these people in place. Again, I'm emphasizing we are not trying to turn everyone into AI ethics experts. It is, it is unreasonable to basically push people through a PhD in Philosophy. What we want to do is to turn everyone into, to create structures where there is an ethics person available for everyone to reach out to. And those people are skilled and literate in ethical decision making, and have access to an expert. So we want to create really a network here.

56:53 And how do we do that? We do that through training. And the training is, again, customized for all of these different layers of knowledge that we want to create, not different layers of skills that we want to create. So for the leadership, we want to have executive education, helping them understand how to think about the ethical aspects of their management decisions. Whereas for the developers or for the procurement team, we want to have different types of training sessions. And we create several different types of trainings. We have webinars, seminars, we have workshops, or sprints. One of the things that we created is a workshop that we call The Mapping, where we take the practitioners, non philosophers, non ethicists, through the experience of making an ethical decision making. And it is very structured. And it is in the form of a floor game, where you really put the participant into the position where they see what they choose, and what is the trade off that they have to engage with. But what we also make sure is that in that whole session, there are, there are times explicitly structured times, where we bring in the expertise of the participant and help them connect with the question at hand.

58:26 So if we are doing this with the developers, and the case is about the, let's say, the ethics of the risk assessment tool, once they know what they are aiming at, what are the value trade-offs that they want to achieve. We ask them to think about how, from their expertise, how could they put it into action. And that's extremely important. Because the whole thing that we are trying to do here is to make sure that ethics is a part of your work, of everyone's work. It is not something outside, it is not that you just ask someone, and if they say go ahead, you go ahead. No, we need, as I showed in the responsible AI, we need everyone's expertise to make sure that the systems run ethical, and they are as ethically as possible. And the roadmap is really where we start actually, because when we engage with a company, we don't know where they are in terms of their AI development. So remember, the first slide that I showed you with the AI ethics trend graphic? Well, you know, AI systems are

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being developed and deployed everywhere. But AI ethics has not been yet put in practice everywhere. So when we see a company or an organization, we don't yet know what is the most pressing, most urgent ethical issue that they are facing. And we want to make sure that we start from there. If you're about to release a product, what we want to do is to go immediately to the analysis and make sure that we can mitigate the risks of that product. If you are more in the position that now you're ready for a strategy making, we can start working on the playbook, on the process and on putting the right people in in places.

1:00:20 If your company is still saying why we need ethics? We probably need to start with training. And in fact training from like, you know, at the level of culture building training, showing why ethics is not, not something that we want to impose on you. But it is what we all, you know, everyone living in this world, we all really care about and we all really need to put in place if we want to make sure that – the systems that treats us in healthcare, that gives us credit in finances, that decides which schools we go to, that determines who can see our CVs when we apply for jobs, that decides how our cars, the transportation system will interact with each other, or how even the pricing is determined for transportation – if we want these things to treat us well, to make sure that we are living in a world where the, where life is good. We all we all have an interest, not just ethicist, but we all have an interest in putting these systems in place. So, if we need to build that culture in a company, then we need to start from the training and we need to build that culture.

1:01:43 So, this is the PiE model. The goal of the, as I said, the point of the PiE model is to make sure that the ethics, the ethical component, is integrated seamlessly, meaningfully, in a robust manner and efficiently. So for that reason, what PiE model proposes is an organizational and a structured integration. It makes, it tries to make sure that ethics questions, ethics problem solving, puzzle-solving is there continuous throughout the lifecycle. And it is multidisciplinary and collaborative with experts. It is explicitly and intentionally multidisciplinary and collaborative. And it is dynamic and responsive. As I mentioned, I joined Northeastern University last year, so the PiE model is also forming the basis of our responsible AI efforts, our responsible AI services at the Institute for Experiential AI at Northeastern University. And what we do is, this is the basic model, but a university and a university like Northeastern has so many resources to offer. So we plug in more capabilities to this model. One of the things that we plug in is tech audits and tech technology consulting. We plugged in an independent advisory board that we offer as a part of the people to the companies when they don't yet have the people. So what more resources we have, and more disciplines we have in our, in our collaboration, we plug in those responsible AI components so that what we have here is as comprehensive and robust, as robust as possible.

1:03:36 And one final word about the puzzle-solving. Why puzzle-solving? Most people talk about problem solving. Or most, you know, ethics is not usually the puzzle-solving, the word puzzle solving is not usually used. Well, because ethical questions are really like puzzles. Dealing with them requires so many moving parts. So you know, we have to define the question first. We have to reveal and uncover the question, we have to understand all the relevant information from all these different disciplines. As we understand that information, we have to figure out what are the underlying ethical tensions, what are the value trade offs that we are dealing with? What is really the problem that we are trying to solve from the ethical perspective? What does it mean to create a system that is ethically better? And how can we determine as we are trying to make a decision, how can we determine all the ethically justifiable options? And what how can we design solutions or mitigations, or at least safeguards, to make sure that whatever we are adding to the world is not only not adding further harm, but in fact tries to prevent some of the harms, some of the risks that are existing in the world?

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1:04:57 And finally, you know they are fascinating. The ethical questions are some of the most important puzzles in our lives. And these questions are fascinating to solve. And we all, as I mentioned, we all have to care because at the core of it, what we are asking is, how can we create systems and societies, a world really, where it is good to live in? Where do we want to live in? So we all have we, we are all stakeholders in this question, in this game. So these are the puzzles that we all have reasons, we are all invested in them. And we have, we have, we all have reasons to want to solve them. And with that, thank you so much. For questions, for staying in touch or for learning more about the lab, here are the relevant links. Thank you.